



Student Name/Number: _____

CARINGBAH HIGH

2020 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- Reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
100

Section I – 10 marks (pages 2-4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5-22)

- Attempt Questions 11– 40
- Allow about 2 hours and 45 minutes for this section

Marker's Use Only							
Section I	Section II					Total	
Q1-10	Q11-17	Q18-23	Q24-30	Q31-36	Q37-40		%
/10	/17	/21	/19	/18	/15	/100	

Section I

10 marks

Attempt Question 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

1) If $\cos \theta = -\frac{12}{13}$ and $180^\circ \leq \theta \leq 360^\circ$, then $\cot \theta =$

A) $-\frac{5}{12}$

B) $-\frac{12}{5}$

C) $\frac{5}{12}$

D) $\frac{12}{5}$

2) What are the asymptotes of the graph of $y = \frac{1}{x^2 - 9}$

A) $x = \pm 3$

B) $x = \pm 9$

C) $y = \pm 3$

D) $y = \pm 9$

3) For the function $f(x) = \frac{x^3}{3} - 5x^2 + 2x + 10$, the gradient is -14 at two points. What are the values of the x -coordinates at these points?

A) $-8, 2$

B) $8, 2$

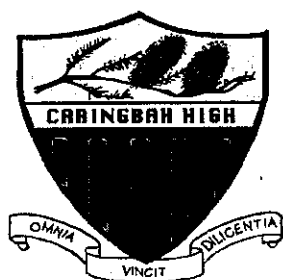
C) $8, -2$

D) $-8, -2$

- 4) What is the domain of the function $f(x) = \sqrt{x} + \frac{1}{\sqrt{2-x}}$?
- A) $(0, 2)$ B) $[0, 2)$
- C) $(0, 2]$ D) $[0, 2]$
- 5) If the z scores on an examination are normally distributed and $P(z \leq N) = 0.6$ for some number N , what is the value of $P(-N \leq z \leq N)$?
- A) 0.1 B) 0.3
- C) 0.2 D) 0.4
- 6) What is the change in amplitude and period when the function $f(x) = \frac{1}{2} \sin 4x$ is transformed into $g(x) = \sin 2x$?
- A) The amplitude is halved and the period is halved B) The amplitude is halved and the period is doubled.
- C) The amplitude is doubled and the period is halved D) The amplitude is doubled and the period is doubled
- 7) Which statement is true for an ungrouped data set with no outliers?
- A) The largest possible range is 2 times the IQR B) The largest possible range is 3 times the IQR
- C) The largest possible range is 4 times the IQR D) The largest possible range is 5 times the IQR

- 8) Which line is perpendicular to $3x + 4y + 7 = 0$?
- A) $4x + 3y - 7 = 0$ B) $3x - 4y + 7 = 0$
- C) $8x - 6y - 7 = 0$ D) $4x - 7y + 7 = 0$
- 9) What is the derivative of 3^{4x+5} ?
- A) $\ln 3 \times 4 \times 3^{4x+5}$ B) $(4x+5) \times 3^{4x+5}$
- C) $\ln 3 \times 3^{4x+5}$ D) $4 \times 3^{4x+5}$
- 10) What is the value of $\ln 2 + \ln 4 + \ln 8 + \dots + \ln 2^{2n}$?
- A) $n^2 \ln 2$ B) $n(n+1) \ln 2$
- C) $n(n+2) \ln 2$ D) $n(2n+1) \ln 2$

End of Section I



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2020 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Advanced Section II Answer Booklet

90 marks

Attempt Questions 11–40

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 11 (2 marks)

Find the values of a and b (in simplified form) such that

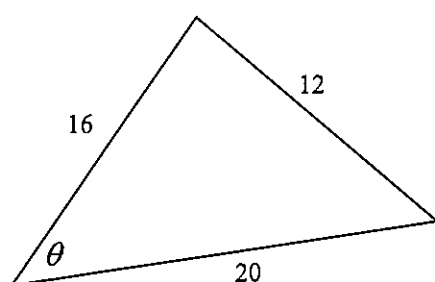
$$\frac{3}{4-\sqrt{2}} = a + \sqrt{b}$$

2

Question 12 (2 marks)

Find the value of θ , correct to the nearest minute

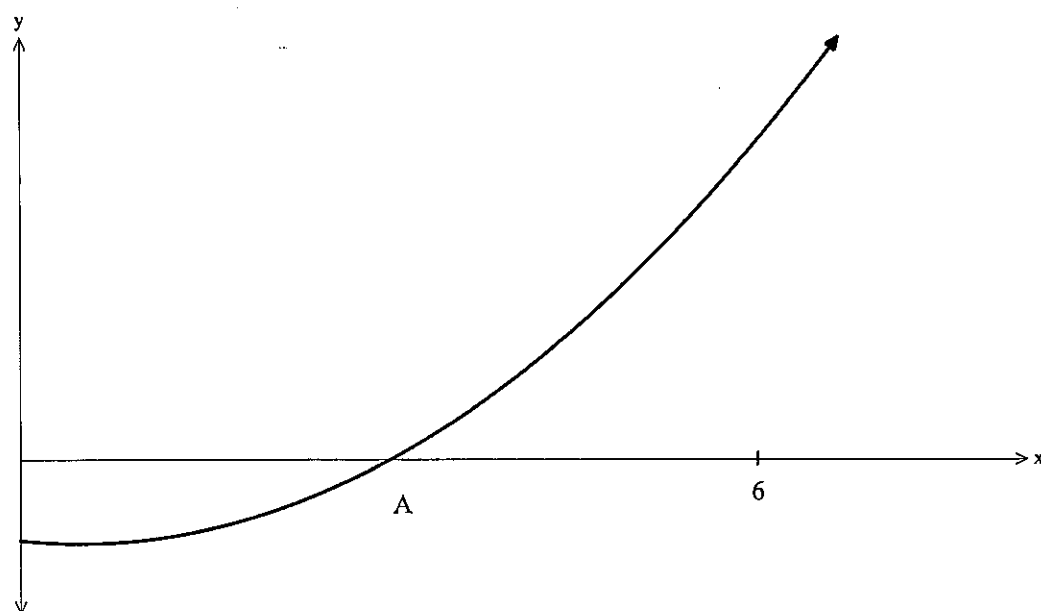
2



NOT TO SCALE

Question 13 (4 marks)

The diagram below shows the graph of $y = x^2 - x - 6$.



(a) What is the coordinate of A?

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(b) Find the area bounded by the x -axis and the curve $y = x^2 - x - 6$ for the interval $0 \leq x \leq 6$.

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Question 14 (3 marks)

Differentiate

(a) $y = x^2 e^x$

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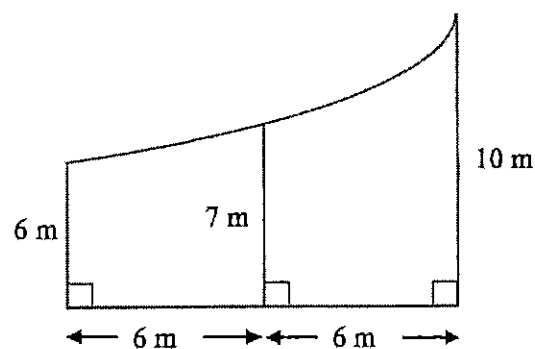
(b) $f(x) = \frac{e^x + 1}{2x}$

2

Question 15 (2 marks)

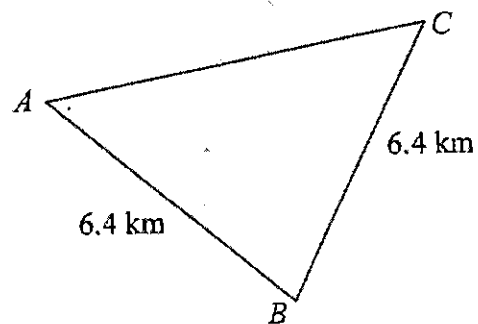
Use two applications of the trapezoidal rule to find an approximation to the area given in the diagram.

2



NOT TO SCALE

Question 16 (2 marks)



NOT TO SCALE

In the diagram, ABC is a triangular airfield with $AB = BC = 6.4$ km. The bearing of B from A is 140° and the bearing of C from B is 032° .

(a) Show that $\angle ABC = 72^\circ$.

1

(b) Find the area of the airfield, correct to the nearest square kilometre.

1

Question 17 (2 marks)

Solve $|2 \cos x - 1| = 1$ for $0 \leq x \leq \pi$

2

Question 18 (6 marks)

Consider the curve $y = 2x^3 - 9x^2 + 12x$.

- (a) Find the coordinates of the stationary points and determine their nature.

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- (b) Show that a point of inflection occurs at $x = \frac{3}{2}$.

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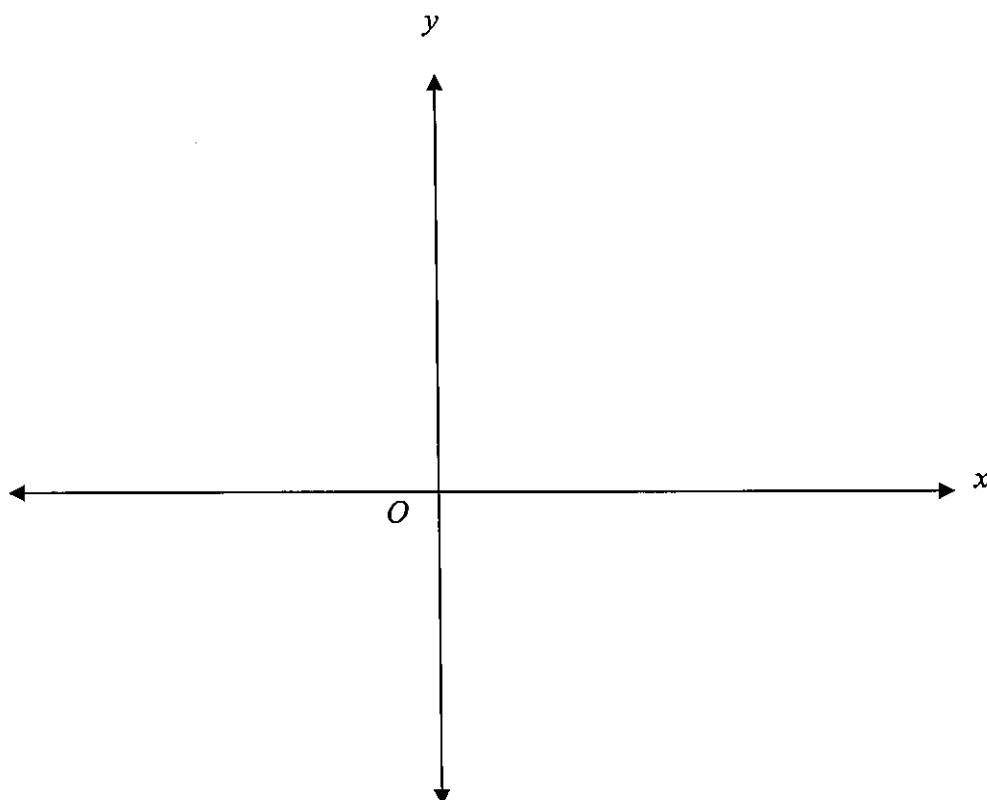
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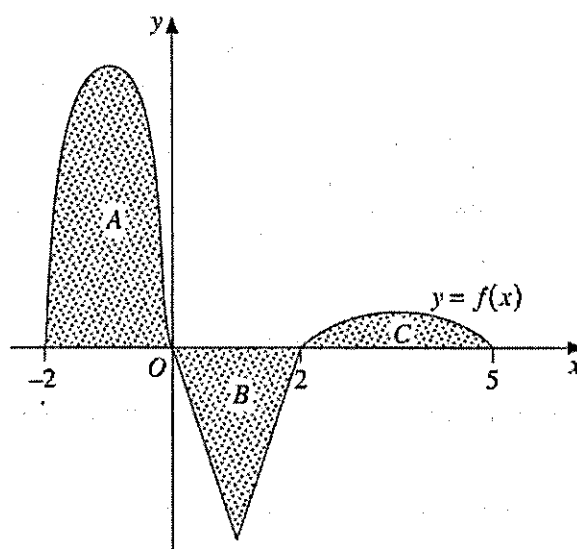
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(c) Sketch the graph $y = 2x^3 - 9x^2 + 12x$, indicating clearly all important features.

2



Question 19 (1 mark)

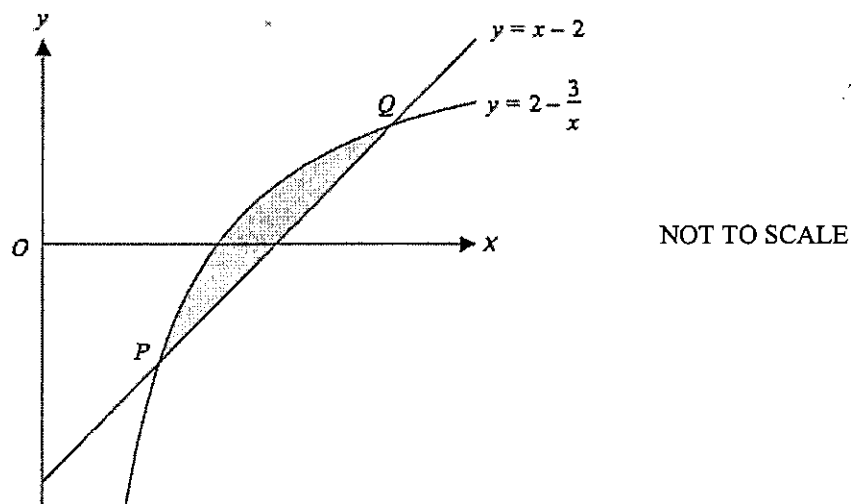


NOT TO SCALE

The graph of the function f is shown in the diagram above. The shaded areas are bounded by $y = f(x)$ and the x axis. The shaded area A is 8 square units, the shaded area B is 3 square units and the shaded area C is 1 square unit.

Evaluate $\int_{-2}^5 f(x) dx$.

Question 20 (5 marks)



The diagram shows the curves $y = 2 - \frac{3}{x}$ and $y = x - 2$ for $x \geq 0$.

- (a) Find the coordinates of the two points P and Q where the two curves intersect. 2

[illegible]

- (b) Hence, find in simplest form, the area of the shaded region contained between the two curves. 3

[illegible]

Question 21 (3 marks)

(a) Show that $\log_x 2 = \frac{1}{\log_2 x}$.

1

(b) Solve the equation $\log_2 x = 4\log_x 2$

2

Question 22 (2 marks)

The completion times for the Oztown triathlon race were normally distributed with mean times 60 minutes and standard deviation 5 minutes. Using the empirical rule, find Ozzie's completion time if he finished ahead of 84% of competitors.

2

Question 23 (4 marks)

The discrete random variable X has a mean of 2 and probability distribution

x	1	2	3	4
$p(x)$	0.3	0.45	a	b

(a) Show that the two equations in terms of a and b are

$$a + b = 0.25$$

$$3a + 4b = 0.8$$

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(b) Hence find the values of a and b .

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Question 24 (2 marks)

Consider the function $f(x) = e^x$ and $g(x) = \ln(x-2)$.

(a) Find the composite function $f(g(x))$.

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(b) Find the interval notation for the range of the composite function.

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Question 25 (2 marks)

If $y = x \sin 2x$, find $\frac{dy}{dx}$

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Question 26 (4 marks)

The table below shows the English marks (x) and the Mathematics marks (y) for a class of 12 students ($A-L$). Only the English mark is available for student L .

	A	B	C	D	E	F	G	H	I	J	K	L
x	67	61	65	67	75	75	69	85	85	89	87	80
y	58	64	66	68	70	72	72	76	80	82	84	

- (a) Calculate the correlation coefficient between x and y for the students A to K .
Describe the nature of the correlation coefficient between x and y .

2

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- (b) Find the equation of the least squares regression line of y on x for the students A to K .
Estimate the Mathematics mark of student L .

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Question 27 (2 marks)

If $y = \frac{e^x}{x+1}$, find $\frac{dy}{dx}$.

2

Question 28 (2 marks)

Find $\int \tan^2 x \, dx$

2

Question 29 (2 marks)

Evaluate $\int_0^2 x(x^2 - 4)^3 \, dx$

2

Question 30 (5 marks)

A metal crate of fixed volume 9 m^3 is to be made in the shape of a rectangular prism with length $2x$ metres, width x metres and height h metres.

- (a) Show that the area $A \text{ m}^2$ of metal required is given by $A = 4x^2 + \frac{27}{x}$. **2**

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- (b) Hence find the minimum area of metal required. **3**

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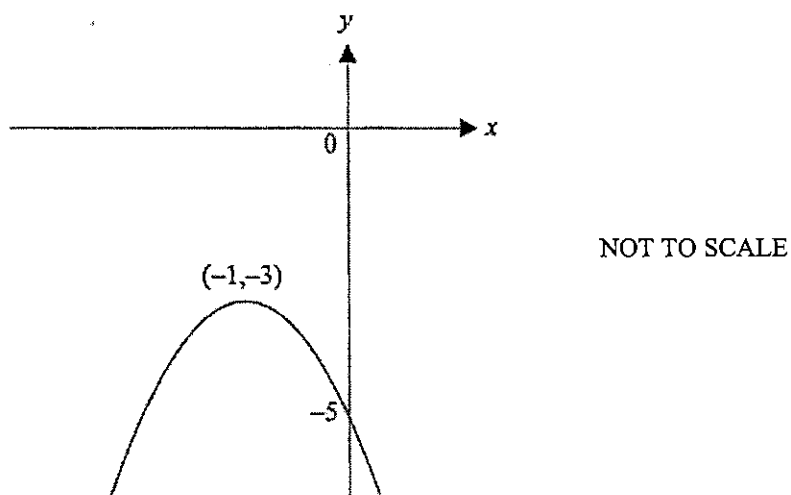
Question 31 (3 marks)

At time (t hours) after 12:00 am, the height (h metres) of the deck of a boat above the level of the jetty is given by $h = 2 \cos\left(\frac{4\pi}{25}t\right) + 1$. Find, correct to the nearest minute, the first time after 12:00 am when the deck of the boat is level with the jetty. 3

[illegible]

Question 32 (3 marks)

The function $f(x) = x^2$ is transformed into a new function whose graph is shown in the diagram below.



Find the equation of the new function in the form $g(x) = k f(x+b) + c$ for some constants k , b and c .

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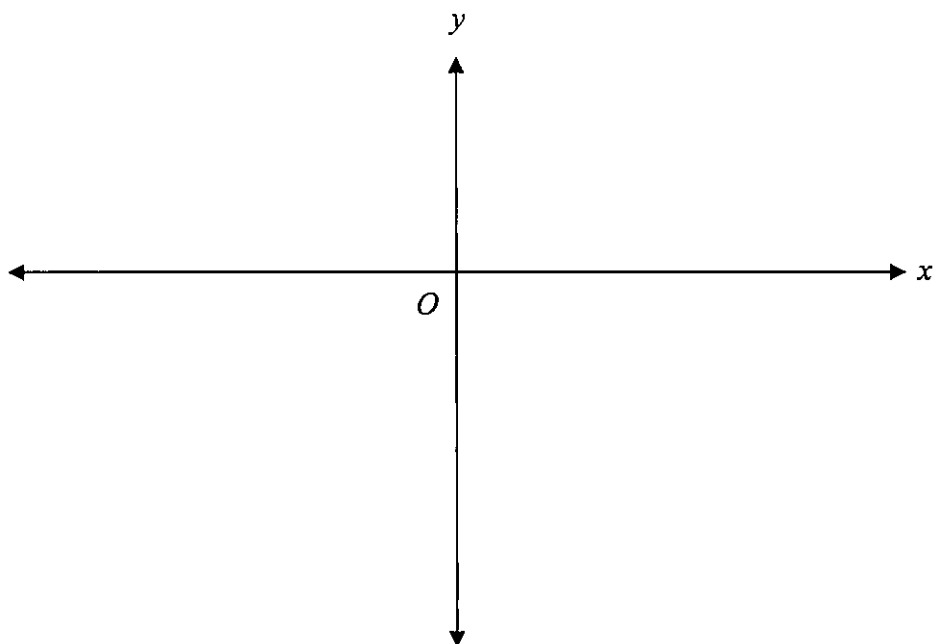
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Question 33 (3 marks)

- (a) On the number plane below, draw the graphs of $y = \cos \pi x$ and $y = 1 - |x|$ for $-3 \leq x \leq 3$.

2



- (b) Hence find the number of solutions of the equation $\cos \pi x = 1 - |x|$ in the domain $(-\infty, \infty)$.

1

Question 34 (3 marks)

If $y = \tan^2 x$, find the values of the constants a and b , such that $\frac{d^2 y}{dx^2} = ay^2 + by + 2$.

3

Question 35 (3 marks)

The continuous random variable X has probability density function $f(x) = \frac{1}{2} \sin x$ for $0 \leq x \leq \pi$.

(a) Find the cumulative distributive function (CDF)

2

(b) Find the first quartile of the distribution.

1

Question 36 (3 marks)

(a) Differentiate $x \log_e x$.

1

(b) Hence or otherwise, evaluate (in exact form), $\int_1^2 \log_e x \, dx$.

2

Question 37 (4 marks)

At time t years after it was purchased the value $\$V$ of a car is given by $V = 25\,000e^{-0.5t}$.

- (a) Find the loss in value of the car during the third year. **1**

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- (b) Find the year in which the car is losing value at a rate of \$100 per year. **2**

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Question 38 (2 marks)

The first term of a geometric series is 16 and the fourth term is $\frac{1}{4}$.

- (a) Find the common ratio. **1**

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- (b) Find the limiting sum of the series. **1**

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Question 39 (5 marks)

A particle is moving in a straight line. At time t seconds it has a displacement x metres from a fixed point O on the line, velocity $v \text{ ms}^{-1}$, and acceleration $a \text{ ms}^{-2}$ is given by $a = 6t - 12$. Initially, the particle is at rest at O .

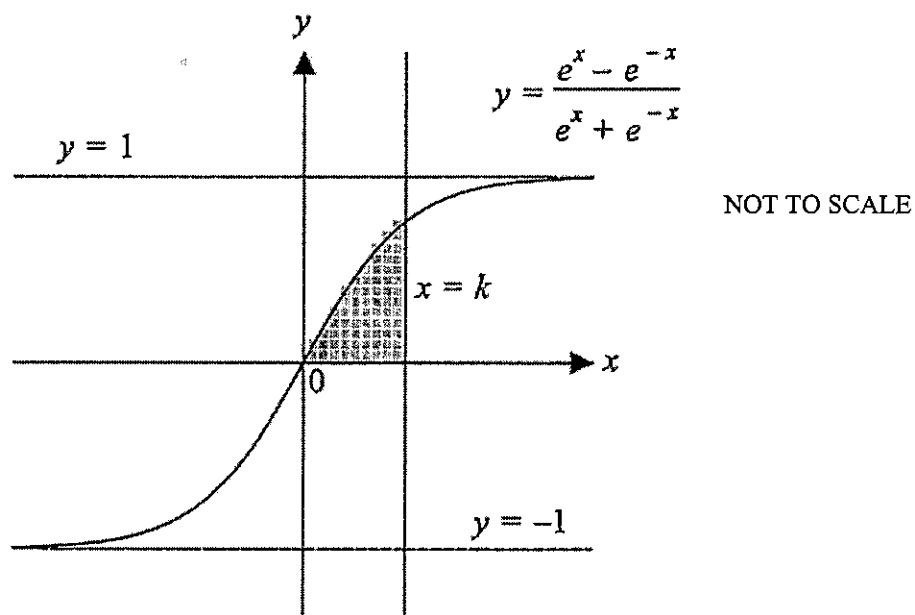
- (a) Find expressions for v and x in terms of t . 3

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- (b) Find when and where the particle is next at rest. 2

[illegible]

Question 40 (5 marks)



The diagram shows the graph of the curve $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

(a) Show that the shaded region bounded by the curve, the x axis and the line

$$x = k, \text{ where } k > 0, \text{ has area } \ln\left(\frac{e^k + e^{-k}}{2}\right).$$

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CARINGBAH HIGH

Student Name/Number:

solutions

2020

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Advanced

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Marker's Use Only							
Section I	Section II					Total	
Q1-10	Q11-17	Q18-23	Q24-30	Q31-36	Q37-40		%
/10	/17	/21	/19	/18	/15	/100	

Section I

10 marks

Attempt Question 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

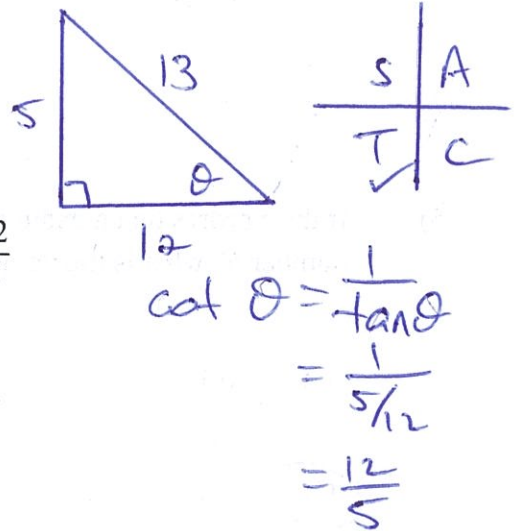
- 1) If $\cos \theta = -\frac{12}{13}$ and $180^\circ \leq \theta \leq 360^\circ$, then $\cot \theta =$

A) $-\frac{5}{12}$

B) $-\frac{12}{5}$

C) $\frac{5}{12}$

D) $\frac{12}{5}$



- 2) What are the asymptotes of the graph of $y = \frac{1}{x^2 - 9}$

A) $x = \pm 3$

B) $x = \pm 9$

C) $y = \pm 3$

D) $y = \pm 9$

$$x^2 - 9 = 0$$

$$(x-3)(x+3) = 0$$

$$x = \pm 3$$

- 3) For the function $f(x) = \frac{x^3}{3} - 5x^2 + 2x + 10$, the gradient is -14 at two points. What are the values of the x -coordinates at these points?

A) $-8, 2$

B) $8, 2$

C) $8, -2$

D) $-8, -2$

$$f'(x) = x^2 - 10x + 2$$

$$x^2 - 10x + 2 = -14$$

$$x^2 - 10x + 16 = 0$$

$$(x-8)(x-2) = 0$$

$$x = 8, 2$$

4) What is the domain of the function $f(x) = \sqrt{x} + \frac{1}{\sqrt{2-x}}$?

$$x \geq 0, x < 2$$

$$0 \leq x < 2$$

$$= [0, 2)$$

A) (0, 2)

☒ B) [0, 2)

C) (0, 2]

D) [0, 2]

5) If the z scores on an examination are normally distributed and $P(z \leq N) = 0.6$ for some number N , what is the value of $P(-N \leq z \leq N)$?

A) 0.1

B) 0.3

☒ C) 0.2

D) 0.4

$$P(-N \leq z \leq N)$$

$$= 2P(0 \leq z \leq N)$$

$$= 2\{P(z \leq N) - 0.5\}$$

$$= 2(0.6 - 0.5)$$

$$= 2(0.1)$$

$$= 0.2$$

6) What is the change in amplitude and period when the function $f(x) = \frac{1}{2} \sin 4x$ is transformed into $g(x) = \sin 2x$?

$$\text{Amplitude} = 1$$

$$\text{Period} = \pi$$

$$\text{Amplitude} = \frac{1}{2}$$

$$\text{Period} = \frac{\pi}{2}$$

A) The amplitude is halved and the period is halved

B) The amplitude is halved and the period is doubled.

C) The amplitude is doubled and the period is halved

☒ D) The amplitude is doubled and the period is doubled

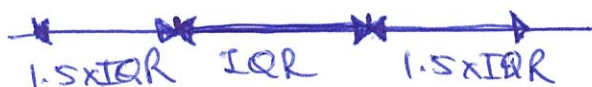
7) Which statement is true for an ungrouped data set with no outliers?

A) The largest possible range is 2 times the IQR

B) The largest possible range is 3 times the IQR

☒ C) The largest possible range is 4 times the IQR

D) The largest possible range is 5 times the IQR



$$(1.5 + 1 + 1.5) \times \text{IQR}$$

$$= 4 \times \text{IQR}$$

8) Which line is perpendicular to $3x + 4y + 7 = 0$?

A) $4x + 3y - 7 = 0$

B) $3x - 4y + 7 = 0$

C) $8x - 6y - 7 = 0$

D) $4x - 7y + 7 = 0$

by $y = 8x - 7$
 $y = \frac{8x}{6} - \frac{7}{6}$
 $y = \frac{4x}{3} - \frac{7}{6}$

$4y = -3x - 7$
 $y = -\frac{3}{4}x - \frac{7}{4}$
 $\therefore m = -\frac{3}{4}$
 $m_{\perp} = \frac{4}{3}$

9) What is the derivative of 3^{4x+5} ?

A) $\ln 3 \times 4 \times 3^{4x+5}$

B) $(4x+5) \times 3^{4x+5}$

C) $\ln 3 \times 3^{4x+5}$

D) $4 \times 3^{4x+5}$

$\frac{d}{dx} 3^{4x+5}$
 $= \ln 3 \times 4 \times 3^{4x+5}$

10) What is the value of $\ln 2 + \ln 4 + \ln 8 + \dots + \ln 2^{2n}$?

A) $n^2 \ln 2$

B) $n(n+1) \ln 2$

C) $n(n+2) \ln 2$

D) $n(2n+1) \ln 2$

$\ln 2 + \ln 2^2 + \ln 2^3 + \dots + \ln 2^{2n}$
 $= \ln 2 + 2\ln 2 + 3\ln 2 + \dots + 2n\ln 2$
 $= (1 + 2 + 3 + \dots + 2n) \ln 2$

$a=1, d=1, n=2n, L=2n$

End of Section I

$S_n = \frac{n}{2}(a+L)$
 $= \frac{2n}{2}(1+2n)$
 $= n(1+2n)$

$\therefore \text{sum} = n(1+2n) \ln 2$



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-

Question 11 (2 marks)Find the values of a and b (in simplified form) such that

$$\frac{3}{4-\sqrt{2}} = a + \sqrt{b}$$

2

$$\frac{3}{4-\sqrt{2}} \times \frac{4+\sqrt{2}}{4+\sqrt{2}}$$

$$= \frac{12+3\sqrt{2}}{16-4}$$

$$= \frac{12+3\sqrt{2}}{12}$$

$$= \frac{6}{6} + \frac{3}{12}\sqrt{2}$$

$$\therefore a = \frac{6}{6}, \sqrt{b} = \frac{3}{12}\sqrt{2}$$

$$b = \left(\frac{3}{12}\sqrt{2}\right)^2$$

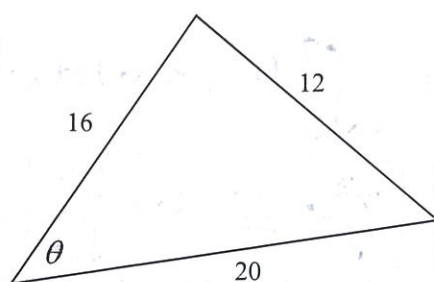
$$= \frac{9 \times 2}{144}$$

$$= \frac{18}{144}$$

$$\therefore b = \frac{9}{98}$$

Question 12 (2 marks)Find the value of θ , correct to the nearest minute

2



NOT TO SCALE

$$\cos \theta = \frac{16^2 + 20^2 - 12^2}{2 \times 16 \times 20}$$

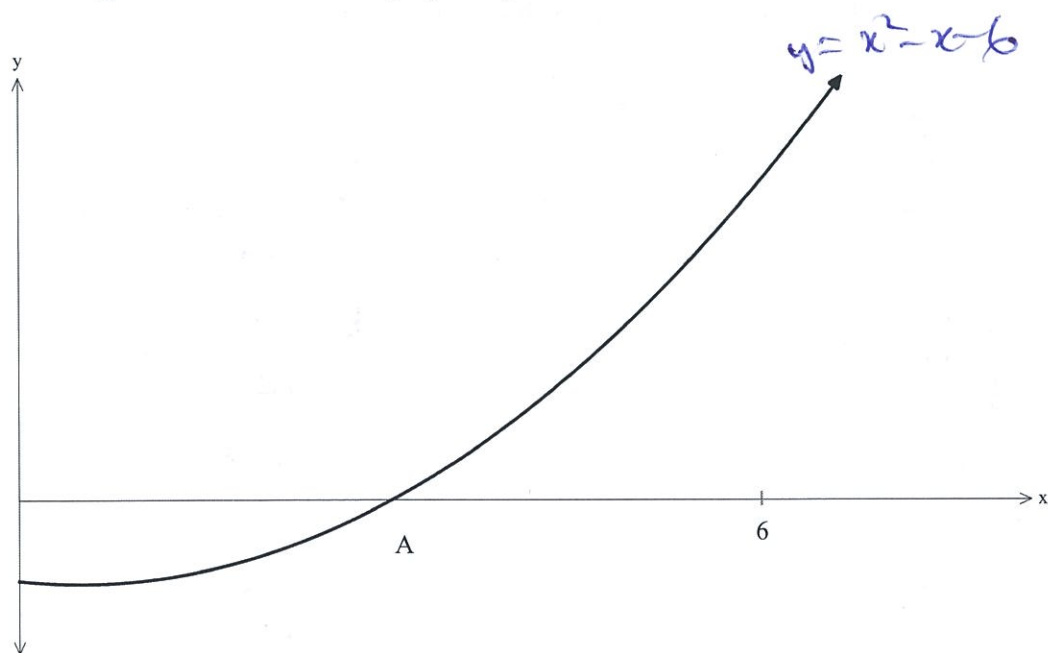
$$= \frac{4}{5}$$

$$\theta = \cos^{-1}\left(\frac{4}{5}\right)$$

$$= 36^\circ 52'$$

Question 13 (4 marks)

The diagram below shows the graph of $y = x^2 - x - 6$.



(a) What is the coordinate of A?

1

$$x^2 - x - 6 = 0.$$

$$(x-3)(x+2) = 0.$$

$$\therefore x = 3, -2.$$

$$\therefore A(3, 0)$$

(b) Find the area bounded by the x -axis and the curve $y = x^2 - x - 6$ for the interval $0 \leq x \leq 6$.

3

$$A = \int_3^6 (x^2 - x - 6) dx + \left| \int_0^3 (x^2 - x - 6) dx \right|$$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} - 6x \right]_3^6 + \left| \left[\frac{x^3}{3} - \frac{x^2}{2} - 6x \right]_0^3 \right|$$

$$= \left(\frac{6^3}{3} - \frac{6^2}{2} - 6(6) \right) - \left(\frac{3^3}{3} - \frac{3^2}{2} - 6(3) \right) + \left| \left(\frac{3^3}{3} - \frac{3^2}{2} - 6(3) \right) - 0 \right|$$

$$= 18 + \frac{27}{2} + \frac{27}{2}$$

$$= 45 \text{ u}^2$$

Question 14 (3 marks)

Differentiate

(a) $y = x^2 e^x$

1

$$y' = x^2 e^x + 2x e^x$$

$$= x e^x (x + 2)$$

(b) $f(x) = \frac{e^x + 1}{2x}$

2

$$f'(x) = \frac{2x(e^x) - (e^x + 1)2}{4x^2}$$

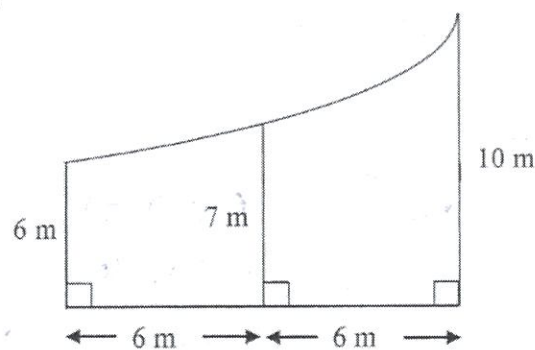
$$= \frac{2xe^x - 2e^x - 2}{4x^2}$$

$$= \frac{xe^x - e^x - 1}{2x^2}$$

Question 15 (2 marks)

Use two applications of the trapezoidal rule to find an approximation to the area given in the diagram.

2

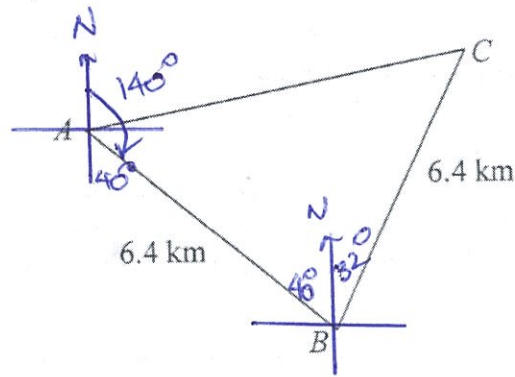


$$A \approx \frac{6}{2} (6 + 2(7) + 10)$$

$$= 3(30)$$

$$= 90 \text{ m}^2$$

Question 16 (2 marks)



NOT TO SCALE

In the diagram, ABC is a triangular airfield with $AB = BC = 6.4$ km. The bearing of B from A is 140° and the bearing of C from B is 032° .

(a) Show that $\angle ABC = 72^\circ$.

1

$$\begin{aligned}\angle ABC &= \angle ABN + \angle NBC \\ &= 40^\circ + 32^\circ \text{ (alt. } \angle\text{s =, || lines)} \\ &= 72^\circ\end{aligned}$$

(b) Find the area of the airfield, correct to the nearest square kilometre.

1

$$\begin{aligned}A &= \frac{1}{2} \times 6.4 \times 6.4 \sin 72^\circ \\ &= 19 \text{ km}^2\end{aligned}$$

Question 17 (2 marks)

Solve $|2 \cos x - 1| = 1$ for $0 \leq x \leq \pi$

2

$$\begin{aligned}2 \cos x - 1 &= 1 & \text{or} & & 2 \cos x - 1 &= -1 \\ 2 \cos x &= 2 & & & 2 \cos x &= 0 \\ \cos x &= 1 & & & \cos x &= 0 \\ \therefore x &= 0 & & & x &= \frac{\pi}{2} \\ & & & & \therefore x &= 0, \frac{\pi}{2}\end{aligned}$$

Question 18 (6 marks)

Consider the curve $y = 2x^3 - 9x^2 + 12x$.

(a) Find the coordinates of the stationary points and determine their nature.

3

$$y' = 6x^2 - 18x + 12$$

$$y' = 0, \text{ stat. pt}$$

$$6x^2 - 18x + 12 = 0$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0$$

$$x = 1, 2$$

$$\text{sub } x=1, y=5$$

$$\text{sub } x=2, y=4$$

$$\therefore (1, 5) \text{ \& } (2, 4)$$

$$y'' = 12x - 18$$

$$\text{sub } x=1, y'' = -6$$

$$< 0, \text{ max}$$

$$\text{sub } x=2, y'' = 6$$

$$> 0, \text{ min}$$

$$\therefore (1, 5) \text{ is a max \& } (2, 4) \text{ is a min stationary point.}$$

(b) Show that a point of inflection occurs at $x = \frac{3}{2}$.

1

$$y'' = 0, \quad 12x - 18 = 0$$

$$12x = 18$$

$$x = \frac{18}{12}$$

$$x = \frac{3}{2}$$

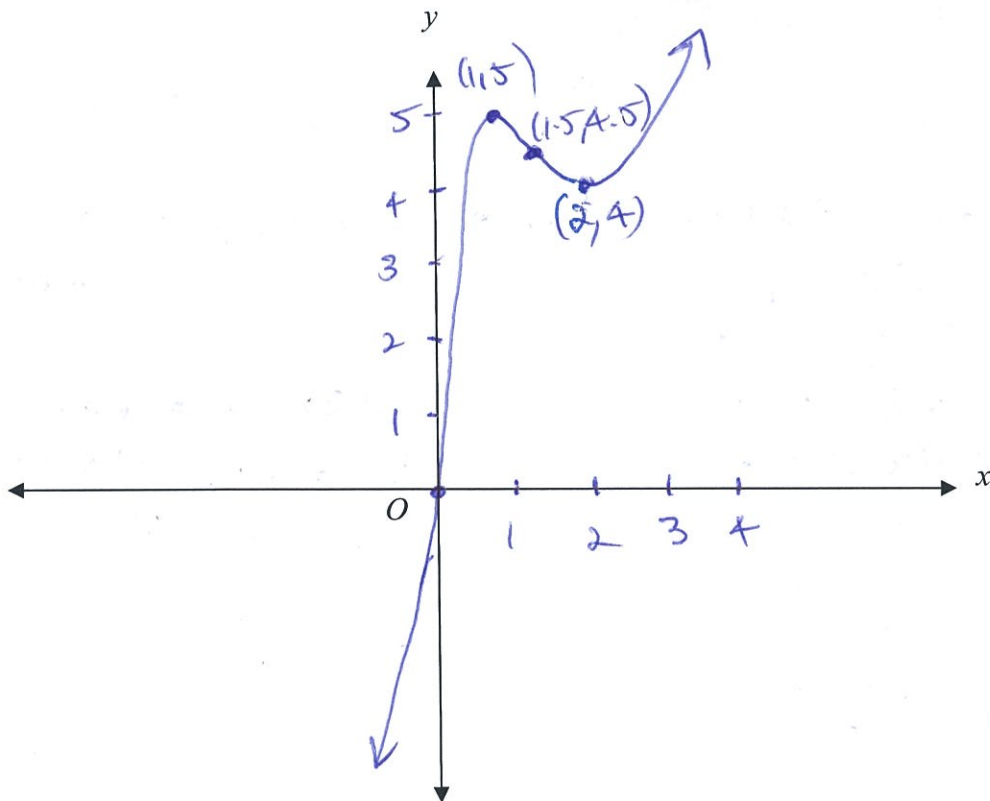
x	1.25	1.5	1.75
y''	-3	0	3

$\therefore$ change in concavity.

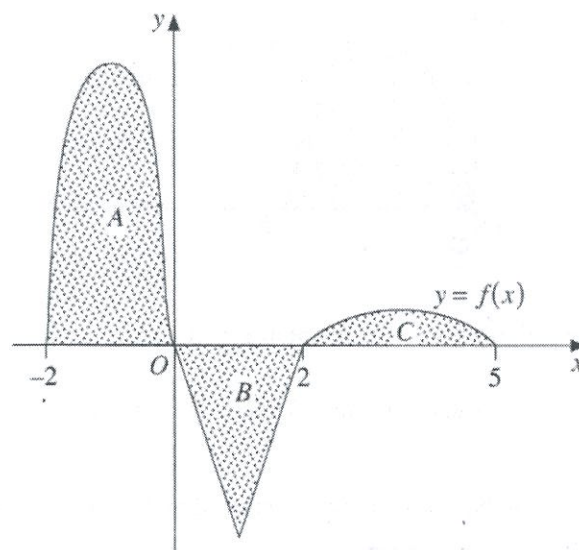
$\therefore$ inflection point occurs at $x = \frac{3}{2}$

(c) Sketch the graph $y = 2x^3 - 9x^2 + 12x$, indicating clearly all important features.

2



Question 19 (1 mark)



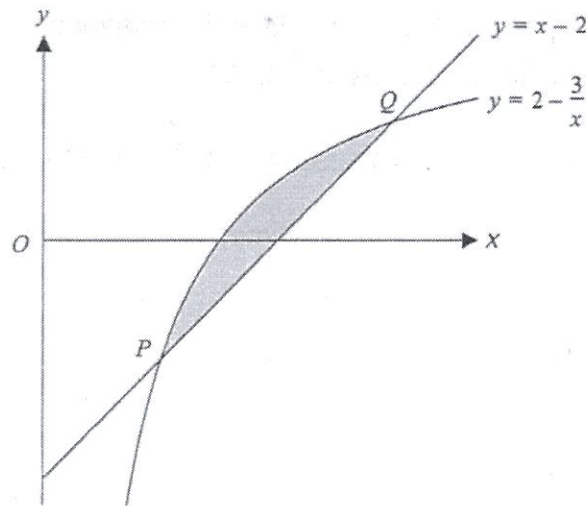
NOT TO SCALE

The graph of the function f is shown in the diagram above. The shaded areas are bounded by $y = f(x)$ and the x axis. The shaded area A is 8 square units, the shaded area B is 3 square units and the shaded area C is 1 square unit.

Evaluate $\int_{-2}^5 f(x) dx$.

$$\int_{-2}^5 f(x) dx = 8 - 3 + 1 = 6 \text{ u}^2$$

Question 20 (5 marks)



The diagram shows the curves $y = 2 - \frac{3}{x}$ and $y = x - 2$ for $x \geq 0$.

- (a) Find the coordinates of the two points P and Q where the two curves intersect. 2

$$2 - \frac{3}{x} = x - 2$$

$$2x - 3 = x^2 - 2x$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 3, 1$$

$$\text{when } x=3, y=1$$

$$\text{when } x=1, y=-1$$

$$\therefore P(1, -1) \text{ and } Q(3, 1)$$

- (b) Hence, find in simplest form, the area of the shaded region contained between the two curves. 3

$$A = \int_1^3 \left[2 - \frac{3}{x} - (x-2) \right] dx$$

$$= \int_1^3 \left(2 - \frac{3}{x} - x + 2 \right) dx$$

$$= \int_1^3 \left(4 - \frac{3}{x} - x \right) dx$$

$$= \left[4x - 3 \ln x - \frac{x^2}{2} \right]_1^3$$

$$= \left(4(3) - 3 \ln 3 - \frac{3^2}{2} \right) - \left(4 - 3 \ln 1 - \frac{1}{2} \right)$$

$$= 12 - 3 \ln 3 - \frac{9}{2} - 4 + \frac{1}{2}$$

$$= (4 - 3 \ln 3) u^2$$

Question 21 (3 marks)

(a) Show that $\log_x 2 = \frac{1}{\log_2 x}$.

1

$$\begin{aligned}\log_x 2 &= \frac{\log_2 2}{\log_2 x} \quad (\text{change of base rule}) \\ &= \frac{1}{\log_2 x}\end{aligned}$$

(b) Solve the equation $\log_2 x = 4 \log_x 2$

2

$$\begin{aligned}\log_2 x &= 4 \times \frac{1}{\log_2 x} \\ \log_2 x &= \frac{4}{\log_2 x}\end{aligned}$$

$$(\log_2 x)^2 = 4$$

$$\log_2 x = \pm 2$$

$$\begin{aligned}\therefore \log_2 x &= 2 & \& \log_2 x &= -2 \\ x &= 2^2 & & x &= 2^{-2} \\ x &= 4 & & x &= \frac{1}{4}\end{aligned}$$

Question 22 (2 marks)

The completion times for the Oztown triathlon race were normally distributed with mean times 60 minutes and standard deviation 5 minutes. Using the empirical rule, find Ozzie's completion time if he finished ahead of 84% of competitors.

2

$$\begin{aligned}50\% + \frac{1}{2} \times 68\% &= 84\% \quad (\text{normal distribution}) \\ 84\% \text{ of the times are } &> \mu - \sigma \\ 60 - 5 &= 55 \text{ minutes}\end{aligned}$$

Question 23 (4 marks)

The discrete random variable X has a mean of 2 and probability distribution

x	1	2	3	4
$p(x)$	0.3	0.45	a	b

(a) Show that the two equations in terms of a and b are

$$a + b = 0.25$$

$$3a + 4b = 0.8$$

2

$$\begin{aligned} 0.3 + 0.45 + a + b &= 1 \\ 0.75 + a + b &= 1 \\ a + b &= 0.25 \end{aligned} \quad \left\{ \begin{aligned} 0.3 + 2(0.45) + 3a + 4b &= 2 \\ 0.3 + 0.9 + 3a + 4b &= 2 \\ 1.2 + 3a + 4b &= 2 \\ 3a + 4b &= 0.8 \end{aligned} \right.$$

(b) Hence find the values of a and b .

2

$$\begin{aligned} 4a + 4b &= 1 \quad \text{--- (1)} \\ -3a + 4b &= 0.8 \quad \text{--- (2)} \\ \hline \text{①} - \text{②} \quad a &= 0.2 \\ \text{sub } a = 0.2 \text{ into (1)} \\ 4(0.2) + 4b &= 1 \\ 0.8 + 4b &= 1 \\ 4b &= 0.2 \\ b &= 0.05 \\ \therefore a &= 0.2, b = 0.05 \end{aligned}$$

Question 24 (2 marks)

Consider the function $f(x) = e^x$ and $g(x) = \ln(x-2)$.

(a) Find the composite function $f(g(x))$.

1

$$\begin{aligned} f(g(x)) &= f(\ln(x-2)) \\ &= e^{\ln(x-2)} \\ &= x-2, \quad x > 2. \end{aligned}$$

(b) Find the interval notation for the range of the composite function.

1

$$(0, +\infty)$$

Question 25 (2 marks)

If $y = x \sin 2x$, find $\frac{dy}{dx}$

2

$$y' = x(2\cos 2x) + \sin 2x(1)$$

$$= 2x\cos 2x + \sin 2x$$

Question 26 (4 marks)

The table below shows the English marks (x) and the Mathematics marks (y) for a class of 12 students ($A-L$). Only the English mark is available for student L .

	A	B	C	D	E	F	G	H	I	J	K	L
x	67	61	65	67	75	75	69	85	85	89	87	80
y	58	64	66	68	70	72	72	76	80	82	84	

- (a) Calculate the correlation coefficient between x and y for the students A to K .
Describe the nature of the correlation coefficient between x and y .

2

$$r = 0.9, \text{ strong positive correlation}$$

- (b) Find the equation of the least squares regression line of y on x for the students A to K .
Estimate the Mathematics mark of student L .

2

$$y = 18 + 0.72x$$

$$\text{when } x = 80, y = 18 + 0.72(80)$$

$$y = 75.6$$

$$\therefore L \div 76$$

Question 27 (2 marks)

If $y = \frac{e^x}{x+1}$, find $\frac{dy}{dx}$.

2

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x+1)e^x - e^x(1)}{(x+1)^2} \\ &= \frac{xe^x + e^x - e^x}{(x+1)^2} \\ &= \frac{xe^x}{(x+1)^2}\end{aligned}$$

Question 28 (2 marks)

Find $\int \tan^2 x \, dx$

2

$$\begin{aligned}\int \tan^2 x \, dx &= \int (\sec^2 x - 1) \, dx \\ &= \tan x - x + c\end{aligned}$$

Question 29 (2 marks)

Evaluate $\int_0^2 x(x^2 - 4)^3 \, dx$

2

$$\begin{aligned}& \frac{1}{2} \int_0^2 2x(x^2 - 4)^3 \, dx \\ &= \frac{1}{2} \left[\frac{(x^2 - 4)^4}{4} \right]_0^2 \\ &= \frac{1}{8} [(x^2 - 4)^4]_0^2 \\ &= \frac{1}{8} [(2^2 - 4)^4 - (0 - 4)^4] \\ &= \frac{1}{8} (0 - 256) \\ &= -32\end{aligned}$$

Question 30 (5 marks)

A metal crate of fixed volume 9 m^3 is to be made in the shape of a rectangular prism with length $2x$ metres, width x metres and height h metres.

(a) Show that the area $A \text{ m}^2$ of metal required is given by $A = 4x^2 + \frac{27}{x}$.

2

$$\begin{aligned} V &= Ah \\ 9 &= 2x^2h \\ h &= \frac{9}{2x^2} \end{aligned} \quad \left\{ \begin{aligned} A &= 2(2x^2) + 2(2xh) + 2(xh) \\ &= 4x^2 + 4xh + 2xh \\ &= 4x^2 + 6xh \end{aligned} \right. \quad \text{sub (1) into (2)}$$

$$\begin{aligned} A &= 4x^2 + 6x \left(\frac{9}{2x^2} \right) \\ &= 4x^2 + \frac{27}{x} \end{aligned}$$

(b) Hence find the minimum area of metal required.

3

$$\begin{aligned} A &= 4x^2 + 27x^{-1} \\ \frac{dA}{dx} &= 8x - 27x^{-2} \\ &= 8x - \frac{27}{x^2} \end{aligned}$$

$$\begin{aligned} \frac{dA}{dx} &= 0, \quad \frac{8x - 27}{x^2} = 0 \\ 8x^3 - 27 &= 0 \\ 8x^3 &= 27 \\ x^3 &= \frac{27}{8} \\ x &= \frac{3}{2} \end{aligned}$$

$$\begin{aligned} \frac{d^2A}{dx^2} &= 8 + 54x^{-3} \\ &= 8 + \frac{54}{x^3} \end{aligned}$$

$$\begin{aligned} \text{sub } x = \frac{3}{2} \quad \frac{d^2A}{dx^2} &= 8 + \frac{54}{(\frac{3}{2})^3} \\ &= 24 \\ &> 0, \text{ min} \end{aligned}$$

$$\begin{aligned} \therefore A_{\min} &= 4 \left(\frac{3}{2} \right)^2 + \frac{27}{(\frac{3}{2})} \\ &= 27 \text{ m}^2 \end{aligned}$$

Question 31 (3 marks)

At time (t hours) after 12:00 am, the height (h metres) of the deck of a boat above the level of the jetty is given by $h = 2 \cos\left(\frac{4\pi}{25}t\right) + 1$. Find, correct to the nearest minute, the first time after 12:00 am when the deck of the boat is level with the jetty. 3

$$h = 2 \cos\left(\frac{4\pi}{25}t\right) + 1$$

when $h=0$,

$$2 \cos\left(\frac{4\pi}{25}t\right) + 1 = 0$$

$$2 \cos\left(\frac{4\pi}{25}t\right) = -1$$

$$\cos\left(\frac{4\pi}{25}t\right) = -\frac{1}{2}$$

$$\frac{4\pi}{25}t = \cos^{-1}\left(-\frac{1}{2}\right)$$

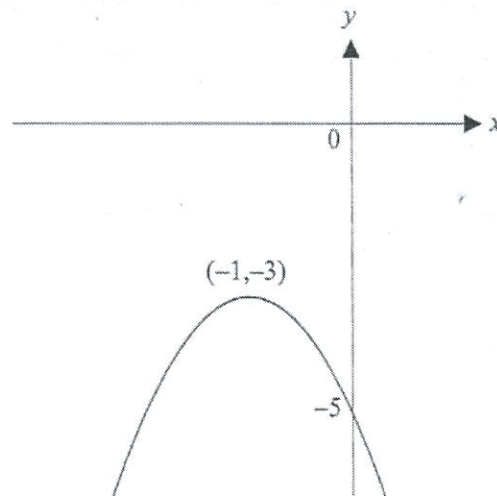
$\sqrt{S}$	A
$\sqrt{T}$	c

$$\frac{4\pi}{25}t = \frac{2\pi}{3}, \frac{4\pi}{3}, \dots$$

$$t = \frac{25}{6}, \frac{25}{3}, \dots$$
$$= 4:10 \text{ am}$$

Question 32 (3 marks)

The function $f(x) = x^2$ is transformed into a new function whose graph is shown in the diagram below.



NOT TO SCALE

Find the equation of the new function in the form $g(x) = k f(x+b) + c$ for some constants k , b and c .

3

The curve is reflected in the x -axis, dilated vertically then translated 1 unit to the left and down by 3 units.

$$\therefore g(x) = k(x+1) - 3$$

$$g(0) = -5$$

$$\therefore k(0+1) - 3 = -5$$

$$k - 3 = -5$$

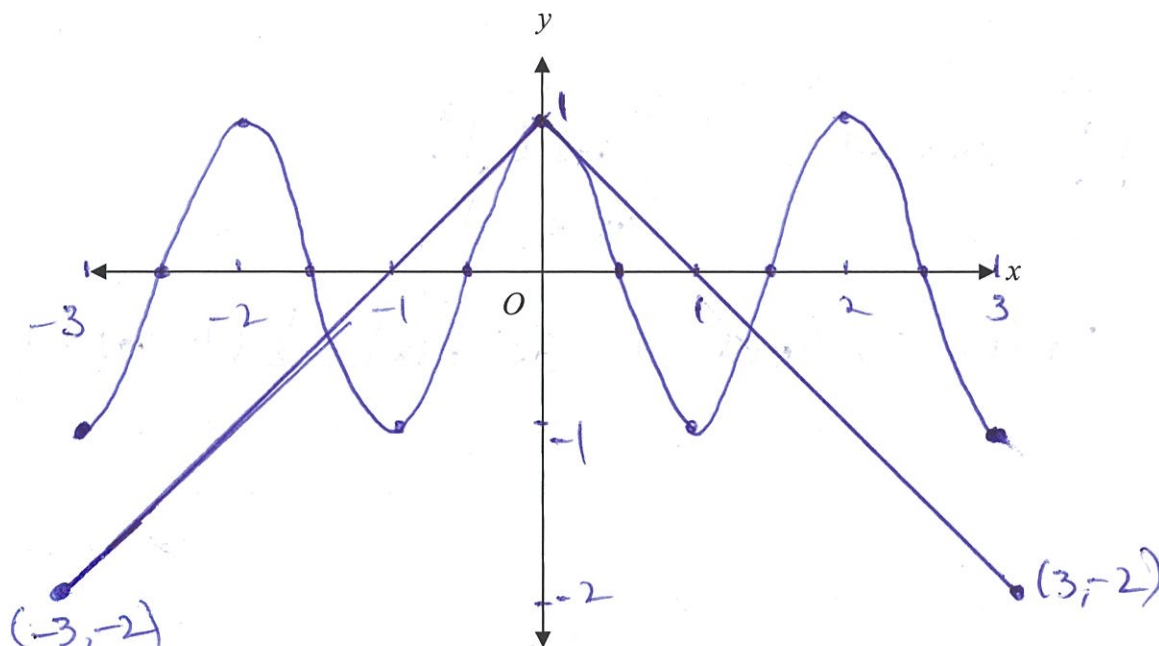
$$k = -2$$

$$\therefore g(x) = -2(x+1)^2 - 3$$

Question 33 (3 marks)

- (a) On the number plane below, draw the graphs of $y = \cos \pi x$ and $y = 1 - |x|$ for $-3 \leq x \leq 3$.

2



- (b) Hence find the number of solutions of the equation $\cos \pi x = 1 - |x|$ in the domain $(-\infty, \infty)$.

1

5 times.

Question 34 (3 marks)

If $y = \tan^2 x$, find the values of the constants a and b , such that $\frac{d^2 y}{dx^2} = ay^2 + by + 2$.

3

$$\begin{aligned}\frac{dy}{dx} &= 2 \tan x \sec^2 x \\ &= 2 \tan x (1 + \tan^2 x) \\ &= 2 \tan x + 2 \tan^3 x\end{aligned}$$

$$\begin{aligned}\frac{d^2 y}{dx^2} &= 2 \sec^2 x + 6 \tan^2 x \sec^2 x \\ &= 2(1 + \tan^2 x) + 6 \tan^2 x (1 + \tan^2 x) \\ &= 2 + 2 \tan^2 x + 6 \tan^2 x + 6 \tan^4 x \\ &= 2 + 8 \tan^2 x + 6 \tan^4 x\end{aligned}$$

sub $y = \tan^2 x$,

$$\frac{d^2 y}{dx^2} = 2 + 8y + 6y^2$$

$$\therefore a = 6, b = 8$$

Question 35 (3 marks)

The continuous random variable X has probability density function $f(x) = \frac{1}{2} \sin x$ for $0 \leq x \leq \pi$.

(a) Find the cumulative distributive function (CDF)

2

$$\int_0^x \frac{1}{2} \sin t \, dt = -\frac{1}{2} [\cos t]_0^x$$

$$= -\frac{1}{2} (\cos x - 1)$$

$$\therefore F(x) = -\frac{1}{2} (\cos x - 1) \quad \text{or} \quad F(x) = \frac{1}{2} (1 - \cos x)$$

(b) Find the first quartile of the distribution.

1

$$F(x) = 0.25, \quad \frac{1}{2} (1 - \cos x) = \frac{1}{4}$$

$$1 - \cos x = \frac{1}{2}$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}$$

Question 36 (3 marks)

(a) Differentiate $x \log_e x$.

1

$$\frac{d}{dx} (x \log_e x) = x \left(\frac{1}{x} \right) + \log_e x (1)$$

$$= 1 + \log_e x$$

$$\therefore \log_e x = \frac{d}{dx} (x \log_e x) - 1$$

(b) Hence or otherwise, evaluate (in exact form), $\int_1^2 \log_e x \, dx$.

2

$$\int_1^2 \log_e x \, dx = [x \log_e x - x]_1^2$$

$$= 2 \log_e 2 - 2 - (\log_e 1 - 1)$$

$$= 2 \log_e 2 - 2 + 1$$

$$= 2 \log_e 2 - 1$$

Question 37 (4 marks)

At time t years after it was purchased the value $\$V$ of a car is given by $V = 25\,000e^{-0.5t}$.

- (a) Find the loss in value of the car during the third year.

1

$$V = 25\,000e^{-0.5t}$$

$$\frac{dV}{dt} = -12\,500e^{-0.5t}$$

- (b) Find the year in which the car is losing value at a rate of $\$100$ per year.

2

$$-12\,500e^{-0.5t} = -100$$

$$12\,500e^{-0.5t} = 100$$

$$e^{-0.5t} = \frac{1}{125}$$

$$\ln e^{-0.5t} = \ln\left(\frac{1}{125}\right)$$

$$-0.5t = \ln 125^{-1}$$

$$-0.5t = -\ln 125$$

$$0.5t = \frac{\ln 125}{0.5}$$

$$t \doteq 9.6566$$

$\therefore$ during the 10th yr.

Question 38 (2 marks)

The first term of a geometric series is 16 and the fourth term is $\frac{1}{4}$.

- (a) Find the common ratio. $a=16, T_4 = \frac{1}{4}$

1

$$T_n = ar^{n-1}$$

$$\frac{1}{4} = 16r^{4-1}$$

$$\frac{1}{4} = 16r^3$$

$$r^3 = \frac{1}{64}$$

$$r = \frac{1}{4}$$

- (b) Find the limiting sum of the series.

1

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{16}{1-\frac{1}{4}}$$

$$= 21\frac{1}{3}$$

Question 39 (5 marks)

A particle is moving in a straight line. At time t seconds it has a displacement x metres from a fixed point O on the line, velocity $v \text{ ms}^{-1}$, and acceleration $a \text{ ms}^{-2}$ is given by $a = 6t - 12$. Initially, the particle is at rest at O .

- (a) Find expressions for v and x in terms of t .

3

$$a = 6t - 12$$

$$v = \frac{6t^2 - 12t + c}{2}$$

$$v = 3t^2 - 12t + c$$

$$\text{when } t=0, v=0 \therefore c=0$$

$$v = 3t^2 - 12t$$

$$x = \frac{3t^3}{3} - \frac{12t^2}{2} + c$$

$$x = t^3 - 6t^2 + c$$

$$\text{when } t=0, x=0 \therefore c=0$$

$$x = t^3 - 6t^2$$

- (b) Find when and where the particle is next at rest.

2

$$\text{At rest, } v=0$$

$$3t^2 - 12t = 0$$

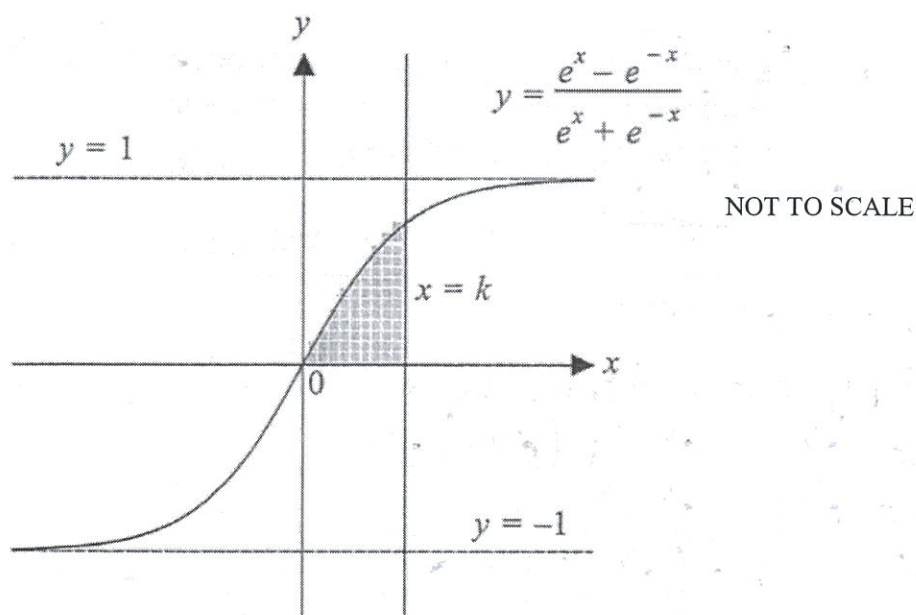
$$3t(t-4) = 0$$

$$t=0, 4$$

$$\text{when } t=4, x = 4^3 - 6(4)^2 = -32$$

$\therefore$ particle next at rest is when $t=4\text{sec}$, when it is 32m to the left of the origin.

Question 40 (5 marks)



The diagram shows the graph of the curve $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

(a) Show that the shaded region bounded by the curve, the x axis and the line

$x = k$, where $k > 0$, has area $\ln\left(\frac{e^k + e^{-k}}{2}\right)$.

2

$$\int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \left[\ln(e^x + e^{-x}) \right]_0^k$$

$$= \ln(e^k + e^{-k}) - \ln(e^0 + e^{-0})$$

$$= \ln(e^k + e^{-k}) - \ln 2$$

$$= \ln\left(\frac{e^k + e^{-k}}{2}\right)$$

- (b) Find, in simplest exact form, the value of k such that the shaded region has area of 1 square unit.

3

$$\ln\left(\frac{e^k + e^{-k}}{2}\right) = 1$$

$$e = \frac{e^k + e^{-k}}{2}$$

$$2e = e^k + e^{-k}$$

Multiply each term by e^k :

$$2e(e^k) = (e^k)^2 + (e^k)(e^k)$$

$$2e(e^k) = (e^k)^2 + 1$$

$$(e^k)^2 - 2e(e^k) + 1 = 0$$

$$\text{let } m = e^k$$

$$m^2 - 2em + 1 = 0.$$

$$m = \frac{2e \pm \sqrt{(-2e)^2 - 4(1)(1)}}{2(1)}$$

$$= \frac{2e \pm \sqrt{4e^2 - 4}}{2}$$

$$= \frac{2e \pm 2\sqrt{e^2 - 1}}{2}$$

$$\therefore m = e \pm \sqrt{e^2 - 1}$$

$$\text{But } m = e^k$$

$$e^k = e \pm \sqrt{e^2 - 1}$$

$$\ln e^k = \ln(e \pm \sqrt{e^2 - 1})$$

$$k = \ln(e \pm \sqrt{e^2 - 1})$$

End of Examination!!!

Section I**10 Marks****Attempt Questions 1-10.****Allow about 15 minutes for this section.**

Select either A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

This page must be handed in with your answer booklet.

	A	B	C	D
1				X
2	X			
3		X		
4		X		
5			X	
6				X
7			X	
8			X	
9	X			
10				X